Running Case (HybridQA): *What was the original name of the city with an AFL club formed in 1989?"* **Target Ground Truth:** "York" • **Modality Fusion:** Table (AFL_Ontario_0) + Wikipedia Bridge (/wiki/Toronto)

Stage 1 (Sec. 3.1): Multi-Source Ingestion

Table: AFL_Ontario_0

Club	Formed	City	height
1989 Hamilton	Toronto Ham. Tigers	Tor. Eagles 1991	

→ Matches constraint: **1989**

Wikipedia Bridge Passage: Toronto → Doc:Toronto

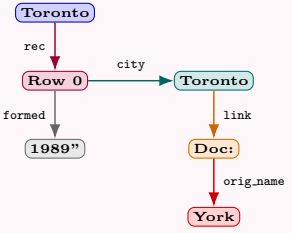
...capital of Ontario. Established in 1793 as to

Retriever Output

• Candidate table rows • Linked Wikipedia passages • Complete context retained

Evidence

Stage 2 (Sec. 3.2): Multi-graph Construction



Star Record Hubs

• Hub binds every cell to its row • Prevents cross-row conflation • Rule-based text links (no LLM)

Guided Reasoning Path

• Best path: Eagles → Row 0 → Toronto → York • Guides model attention (no filtering) • Compact serialized prompt

Graph + Path

Stage 3 (Sec. 3.3): Two-Tier Controlled Engine

Tier 1: Primary Reader [LLM Call 1]

• **Prompt Packaging**: Compressed graph (deduplicated passages + row blocks) + guided reading path • **Controlled Decision**:

- Clear evidence → **Fast Exit Answer** (1 call total)
- Ambiguity or missing facts → **Diagnostic Handoff**

Diagnostic Handoff: {"focus": ["Row 0", "Row 1"], "check": "original name"}

Targeted Bridge Retrieval [0 LLM Calls]: Fetches missing passages for disputed rows; builds focused capsule

Tier 2: Targeted Reader [LLM Call 2]

• **Prompt Packaging**: Transmits *only missing & disputed facts* (discards all unrelated rows) • Resolves candidate ambiguity via verified facts (≤ 2 calls total)

Fast Exit

Tier 2 Verified

Stage 4 (Sec. 3.4): Grounded Provenance

Verified Answer Span

"York"

• Exact context-grounded extraction • Zero hallucination

Auditable Provenance

• Table: AFL_Ontario_0 • Row: Row 0 (Tor. Eagles) • Cell: City: Toronto • Link: /wiki/Toronto • Text: *...town of York...*"

Reverse Grounding Fallback

• Maps answer back to graph coordinates • Direct synthesis if disconnected • Auditable evidence trail